MRI GRADIENT SIGNAL CHAIN

From Pulse Sequence to Spatial Encoding

GE MRI

How gradient systems work – where problems occur – and how they affect the image.

PULSE SEQUENCE → WAVEFORM GENERATION → GRADIENT CONTROLLER → GPA (X,Y,Z) → GRADIENT COIL → SPATIAL ENCODING → IMAGE

WHY GRADIENTS MATTER

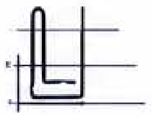
- Gradients encode spatial location of the MR signal.
- They turn frequency and phase into position.
- The three axes (X, Y, Z) work together to fill k-space.
- Any error in the gradient system shows up as an image artifact.

KEY GRADIENT SPECIFICATIONS

- Amplitude (Strength): mT/m
- Slew Rate: T/m/s
- Rise Time: ms (10–90%)
- Duty Cycle / Cooling: %
- Linearity: ppm (parts per million)
- Eddy Current Performance
- Acoustic Noise / Vibration
- PNS (Peripheral Nerve Stimulation) Limits

1. THE GRADIENT SYSTEM SIGNAL CHAIN – OVERVIEW

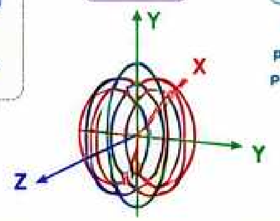
Pulse Sequence (Scanner) → Waveform Generator (Sequence Computer) → Gradient Controller (Real-time Interface) → Gradient Power Amplifiers (GPA): GPA X (Axial), GPA Y (Coronal), GPA Z (Sagittal) → Gradient Coils (X, Y, Z) → Spatial Encoding (k-space) → Image (Reconstruction)



Waveform Generator: Creates gradient waveforms (Gx, Gy, Gz) in time.

Gradient Controller: Formats, scales and sends commands to GPA.

GPA: High-current amplifiers that drive the gradient coils with the commanded current.



Spatial Encoding: Frequency & phase encoding produce k-space data.

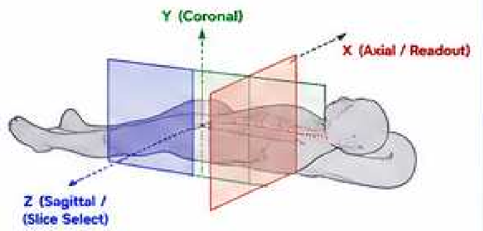
Image: k-space is reconstructed into the final image.

2. HOW X, Y & Z GRADIENTS CREATE AN IMAGE

ROLES OF THE GRADIENTS

Z Gradient (Slice Selection)	Applied during RF pulse. Selects a slice by making the Larmor frequency vary along Z.
Y Gradient (Phase Encoding)	Applied for a short time before readout. Adds a phase offset that varies with position along Y.
X Gradient (Frequency / Readout)	Applied during readout. Causes signal frequency to vary along X. The ADC samples this.

3D VIEW – ORIENTATION OF AXES



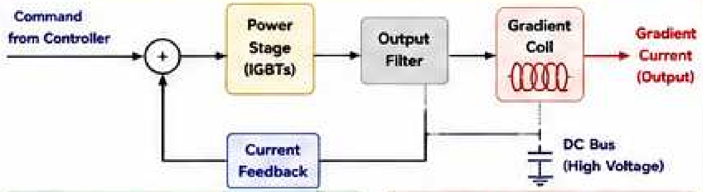
Right-Hand Coordinate System
Thumb: X Index: Y Middle: Z

3. GRADIENT WAVEFORMS – EXAMPLES (X AXIS)

Function	Waveform (Ideal)	Purpose
Slice Select (Gz)	RF	Select slice thickness.
Phase Encode (Gy)		Imparts phase variation.
Readout (Gx)	RO	Encodes frequency (left ↔ right).
Spoiler / Crusher		Dephases residual transverse magnetization.

Real waveforms are not ideal – they have finite rise time and system delays.

4. GRADIENT POWER AMPLIFIER (GPA) – BLOCK DIAGRAM



GPA FUNCTIONS

- Amplify command to high current (up to 800 A peak)
- Provide fast, accurate current control
- Protect against overcurrent, overvoltage, overtemp
- Monitor and report faults
- Support all three axes independently

COMMON GPA FAULTS

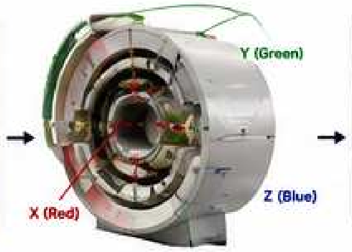
- Overcurrent
- Overvoltage
- Overtemperature
- Short to ground / DC offset
- Amplifier hardware failure

5. WHERE PROBLEMS OCCUR IN THE CHAIN

Stage	What Can Go Wrong	What You May See (Image / System)	How to Check
Waveform Generation	Incorrect scaling, timing, truncation	Wrong FOV, distortion, shift, incomplete k-space	Protocol review, logs, comparison scan
Gradient Controller	Communication error, scaling, timing jitter	Intermittent artifacts, dropouts, wrong gradients	Error logs, oscilloscope, self test
GPA (X/Y/Z)	Overcurrent, overheating, amplifier fault	System errors, no image, axis not working	Service Browser logs, GPA diagnostics
Gradient Coil	Open/short, insulation fault, loose connection	Axis offline, high noise, unusual heating	Coil resistance/inductance, insulation test
Cables / Connectors	Loose, corroded, intermittent	Intermittent artifacts, axis errors	Visual inspect, wiggle test, continuity test
System / Cooling	High temp, low flow, fans, filters	Thermal faults, reduced performance	Temp logs, flow, fans, water quality

6. GRADIENT HARDWARE PATH

GPA Cabinets (X, Y, Z)



Gradient Coil Assembly Inside Magnet

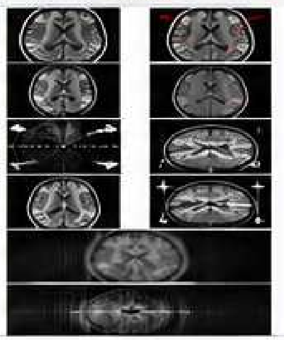
Cooling System (Pumps, Heat Exchanger, Filters, Hoses)

Gradients are water-cooled. Proper cooling is critical for performance and longevity.

7. COMMON IMAGE ARTIFACTS FROM GRADIENT PROBLEMS

Artifact (What You See)	Likely Cause	Which Axis Usually Involved
Geometric Distortion / Warping	Gradient nonlinearity, eddy currents	Usually Readout (X)
Image Shift	Gradient offset (DC), incorrect timing	Any Axis
N/2 Ghosting (EPI)	Polarity mismatch, timing error	Phase Encode (Y)
Scaling Error (Wrong FOV)	Amplitude calibration error	Any Axis
Blurring / Poor Resolution	Low amplitude, long rise time	Usually Readout (X)
Directional Streaking	Eddy currents, poor pre-emphasis	Along the Readout Axis

Example



8. GRADIENT TROUBLESHOOTING APPROACH

1. Review the image and recognize the pattern.
2. Check the system and gradient status for any errors.
3. Determine which axis is likely involved.
4. Change orientation (swap freq/phase) and see if the artifact follows the axis.
5. Run prescan / coil check / system baseline.
6. Check gradient calibration (offset, gain, linearity, delay).
7. Use test tools (impulse, ramp, PNS, dB/dt monitor).
8. Isolate the stage (controller → GPA → coil → cables).

GOOD IMAGES REQUIRE

- Correct waveforms
- Accurate current
- Fast response
- Linear fields
- Proper timing
- Effective eddy current compensation
- Stable temperature and cooling

When gradients are accurate in time, amplitude and linearity – the image is sharp, undistorted and reliable.

ENGINEER NOTES

- Always document changes and results.
- Compare with baseline and other coils/axis.
- Escalate if data is inconclusive.

ABBREVIATIONS

GPA	=	Gradient Power Amplifier	RO	=	Readout	dB/dt	=	Rate of Change of Gradient Field
FOV	=	Field Of View	PNS	=	Peripheral Nerve Stimulation	EPI	=	Echo Planar Imaging
PE	=	Phase Encode	DC	=	Direct Current	ADC	=	Analog to Digital Converter

SAFETY REMINDER

- High currents and voltages are present in GPA.
- Allow time for discharge. Follow lockout/tagout.
- Verify cooling. Hot gradients can fail quickly.
- Follow all GE safety procedures.

THE BOTTOM LINE

Gradients encode space.
Good gradients make great images.
Understand the chain. Verify the system. Fix the cause.

GE HealthCare
Imagination at work